

CSC633 MACHINE LEARNING



TEXAS A&M UNIVERSITY

Department of Computer
Science & Engineering

CSCE 633: Machine Learning

Lecture 1-3: Examples, Models, and Accuracy

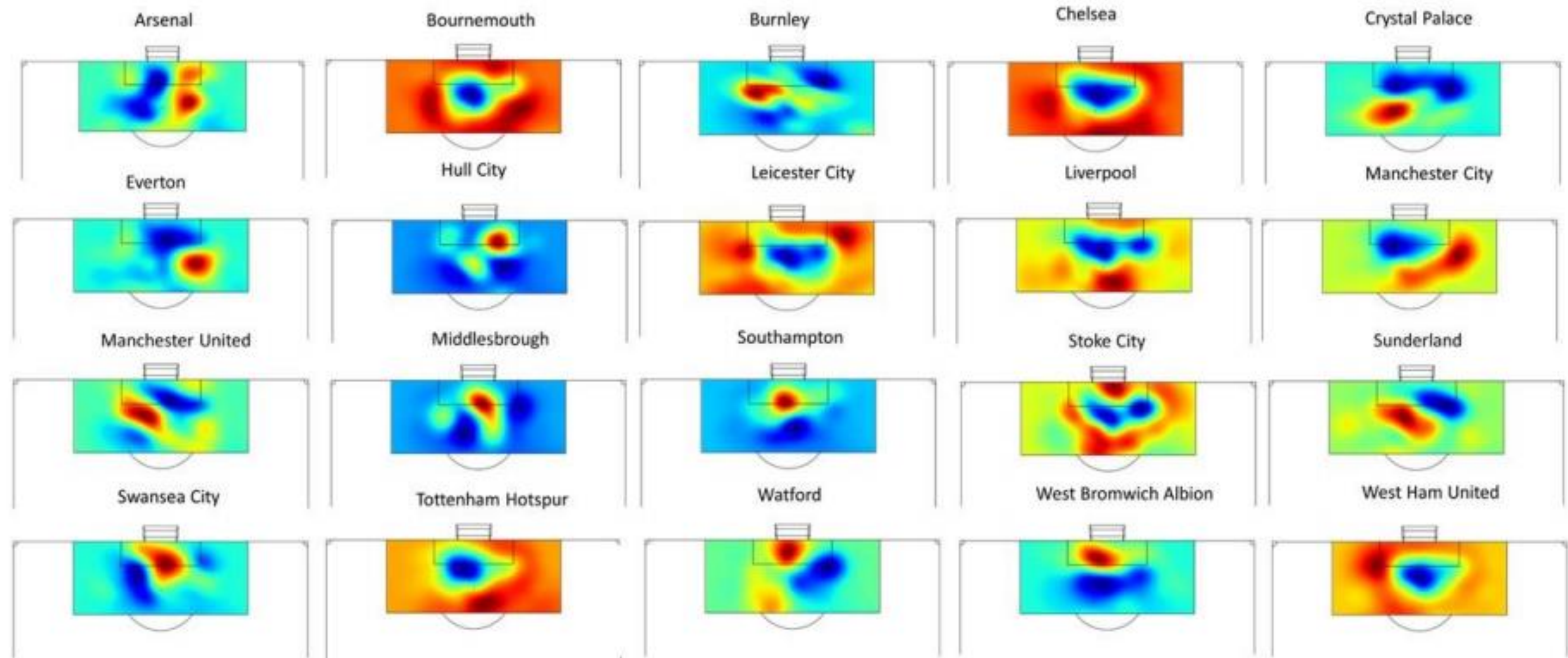
Texas A&M University

Bobak Mortazavi

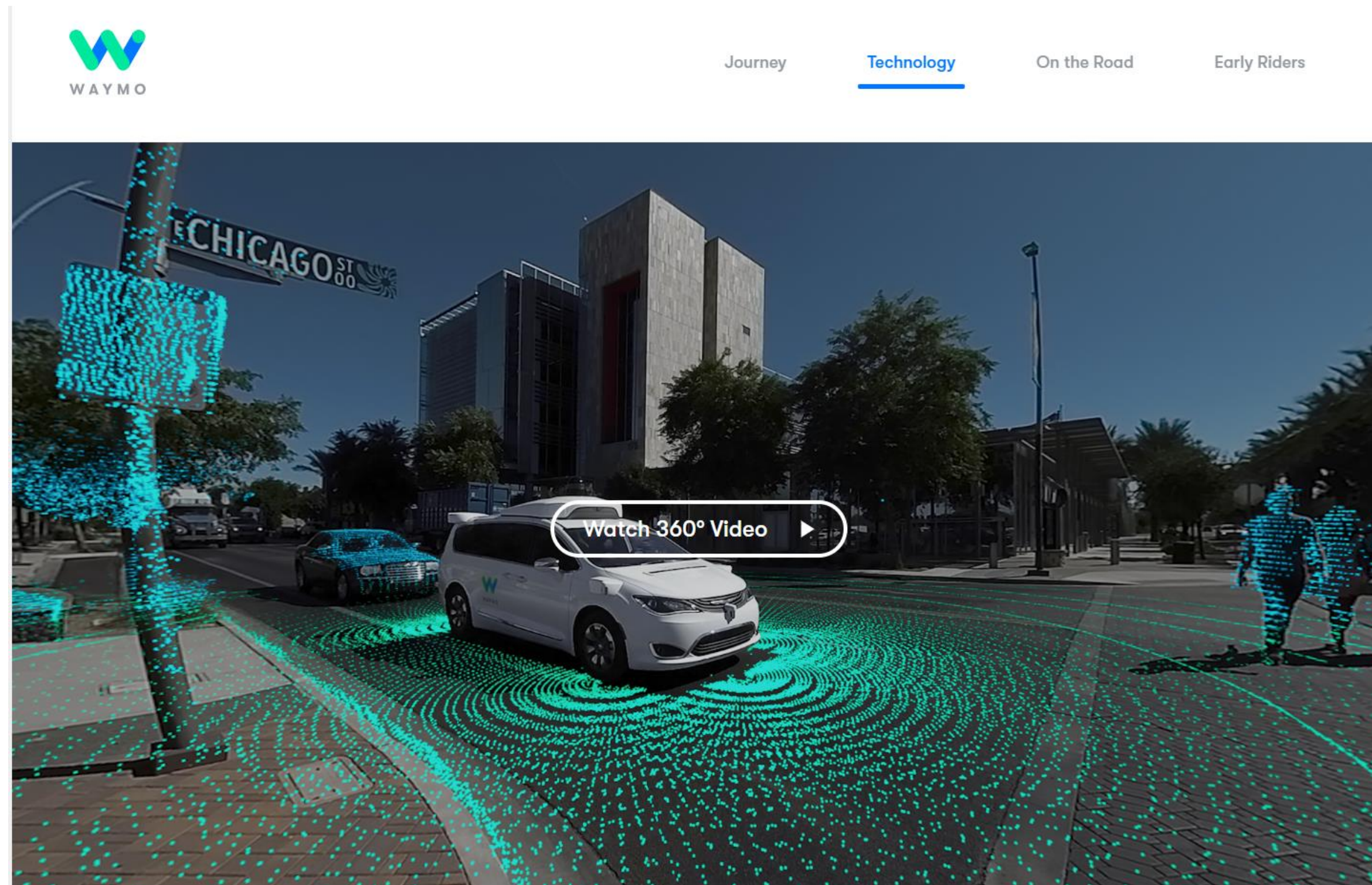
Topics

- Introduction to Machine Learning
 - What is machine learning?
 - Models and accuracy
 - Some general supervised learning examples

Some other examples of machine learning!



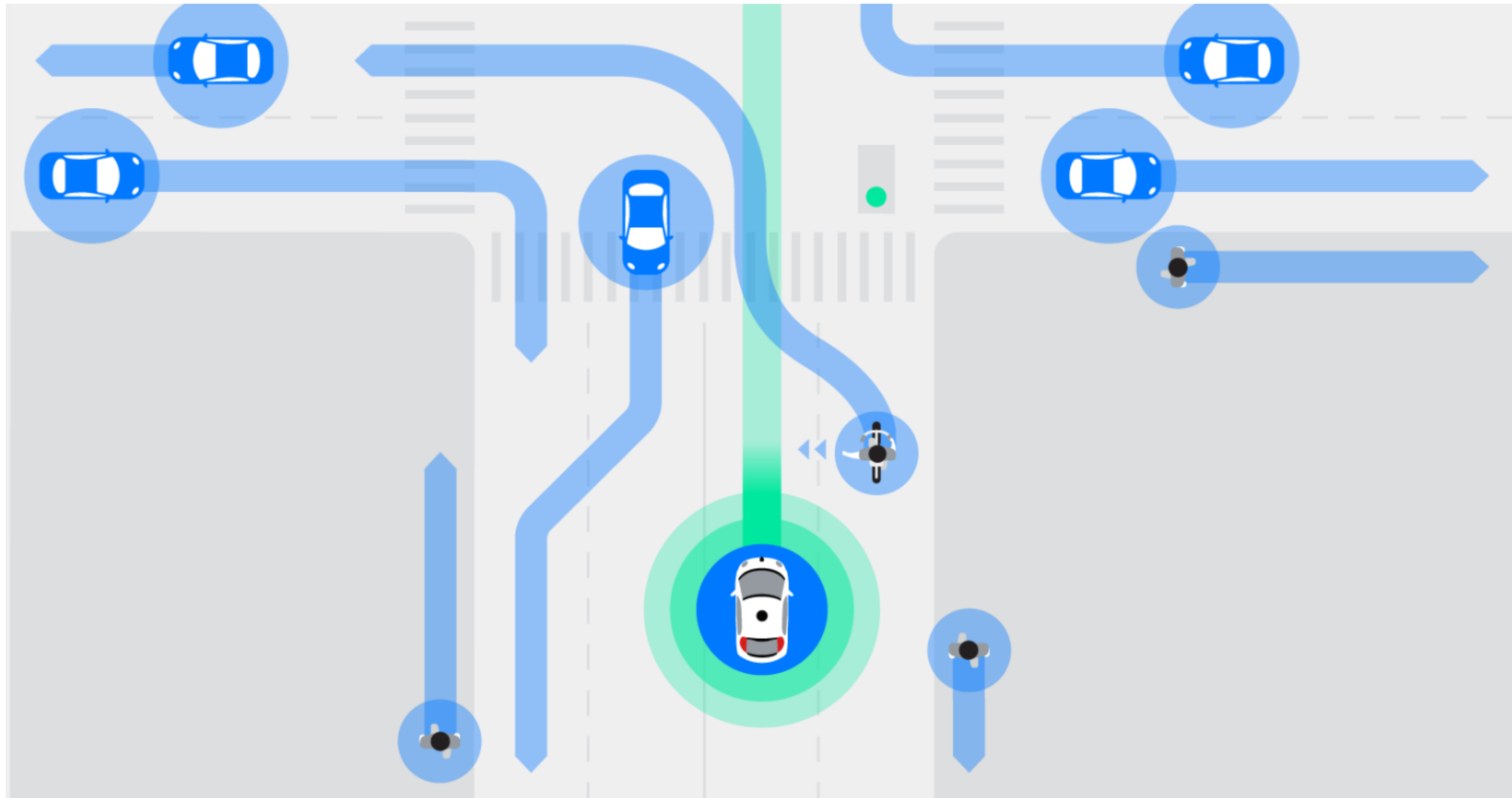
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Self Driving Challenges (1)



Self Driving Challenges (2)



Advertising Example (1)

- Assume we are trying to improve sales of a product through advertising in three different markets:
TV Newspaper Online/Mobile
- Then we can define the advertising budget as:
 - **Input variables** x_i : Consisting of TV budget $x_{i,1}$, newspaper budget $x_{i,2}$, and mobile budget $x_{i,3}$ (also known as the **independent** variables)
 - **Output variable** y : sales numbers (also known as the **response** variable or **dependent** variable)
 - Can generalize $x_{i,1}, x_{i,2}, \dots, x_{i,m} = x_i$

Advertising Example (2)

- Assume we are trying to improve sales of a product through advertising in three different markets:
TV Newspaper Online/Mobile
- Then we can define the advertising budget as:
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 - We then seek to model the relationship $Y = f(X) + \varepsilon$ where f is the systemic information and ε is random error

Advertising Example (3)

- Assume we are trying to improve sales of a product through advertising in three different markets:
TV Newspaper Online/Mobile
- Then we can define the advertising budget as:
 - **Input variables** x_i : Consisting of TV budget $x_{i,1}$, newspaper budget $x_{i,2}$, and mobile budget $x_{i,3}$ (also known as the **independent** variables)
 - **Output variable** y : sales numbers (also known as the **response** variable or **dependent** variable)
 - Can generalize $x_{i,1}, x_{i,2}, \dots, x_{i,m} = x_i$
 - We then seek to model the relationship $Y = f(X) + \varepsilon$ where f is the systemic information and ε is random error
 - **How do we learn f ?**

Why estimate f ? (and how?) (1)

- Prediction
- Inference

Why estimate f ? (and how?) (2)

- Prediction

In many situations, x_i is available to us but y is not. We would like to estimate y as accurately as possible

- Inference

Why estimate f ? (and how?) (3)

- Prediction

In many situations, x_i is available to us but y is not. We would like to estimate y as accurately as possible

- Inference

In some situations both x_i and y are available and we would like to understand how y is affected by changes in x_i

Predicting f ? (1)

- Seek to learn an $\hat{f}(x)$ that generates a $\hat{y}$, which estimates y as closely as possible by attempting to minimize the **reducible error**.
- We have to accept that there may always be **irreducible error** that is independent of x . Why might that be?

Predicting f? (2)

- Seek to learn an $\hat{f}(x)$ that generates a $\hat{y}$, which estimates y as closely as possible by attempting to minimize the **reducible error**.
- We have to accept that there may always be **irreducible error** that is independent of x . Why might that be?
- We can calculate the **expected value** of the average error as:

$$\mathbb{E}(y - \hat{y}) = \mathbb{E}[f(x) + \varepsilon - \hat{f}(x)]^2 = [f(x) - \hat{f}(x)]^2 + \text{Var}(\varepsilon)$$

- The focus of this class is to learn the various techniques that estimate f and minimize this reducible error.

So how do we estimate f ? (1)

- We need **training data** to learn from

So how do we estimate f? (2)

- We need **training data** to learn from
- Define training samples $S = (x_1, y_1), \dots, (x_n, y_n)$ where

$$x_i = (x_{i,1}, x_{i,2}, \dots, x_{i,p})^T$$

for n subjects and p dimensions.

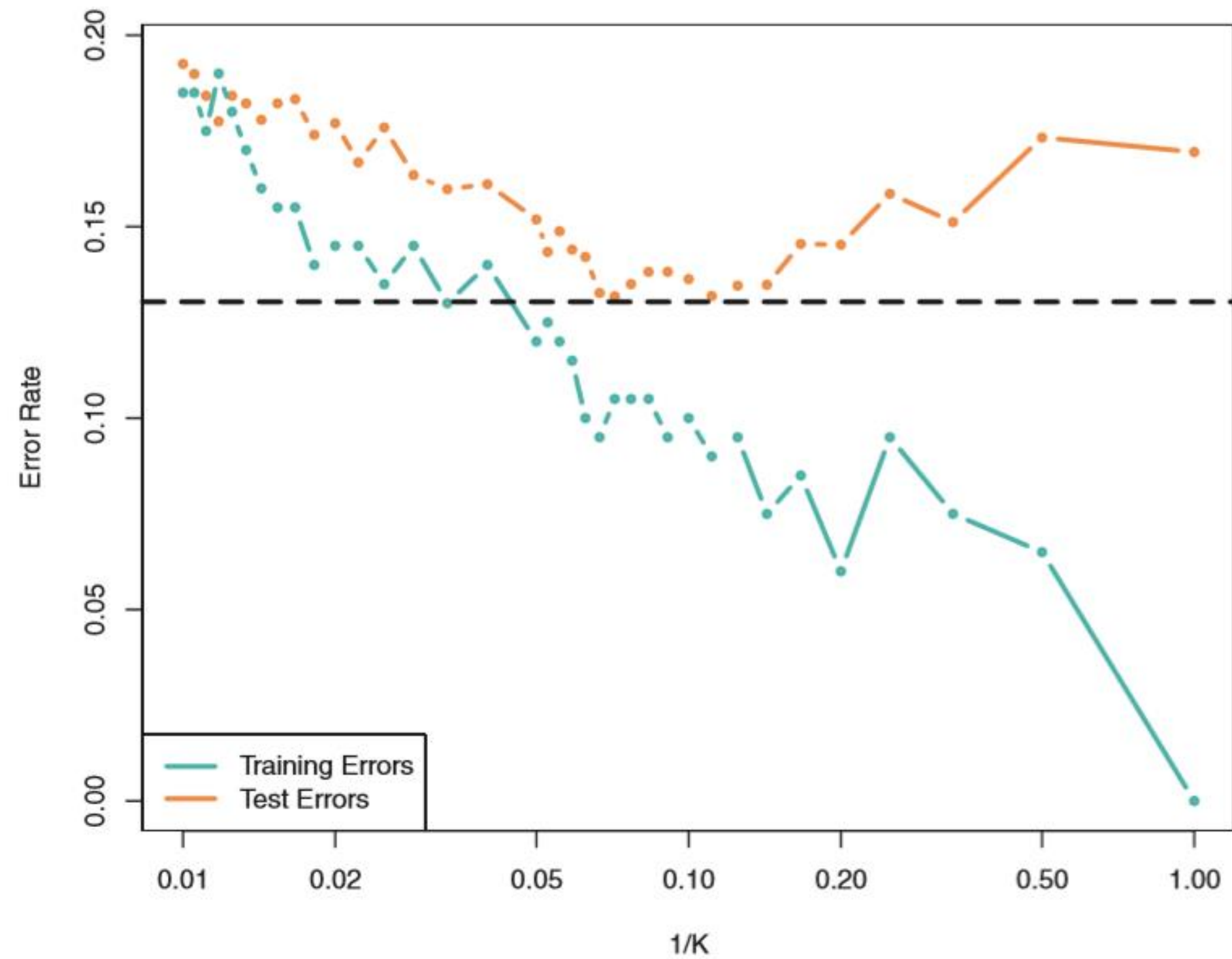
So how do we estimate f? (3)

- We need **training data** to learn from
- Define training samples $S = (x_1, y_1), \dots, (x_n, y_n)$ where
$$x_i = (x_{i,1}, x_{i,2}, \dots, x_{i,p})^T$$
for n subjects and p dimensions.
- Where y can be:
 - Categorical or nominal – the problem we solve is classification or pattern recognition
 - Continuous – the problem we solve is regression
 - Categorical with order, or ordinal (example: Grades A, B, C, D, F) – the problem we can solve is ordinal regression

So how do we estimate f? (4)

- We need **training data** to learn from
- Define training samples $S = (x_1, y_1), \dots, (x_n, y_n)$ where
$$x_i = (x_{i,1}, x_{i,2}, \dots, x_{i,p})^T$$
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- Where y can be:
 - Categorical or nominal – the problem we solve is classification or pattern recognition
 - Continuous – the problem we solve is regression
 - Categorical with order, or ordinal (example: Grades A, B, C, D, F) – the problem we can solve is ordinal regression
- **Please Note! NOTATIONS MAY DIFFER ACROSS BOOKS**

Things to watch out for!



Takeaways

- Some general supervised learning examples
- Understanding key notation